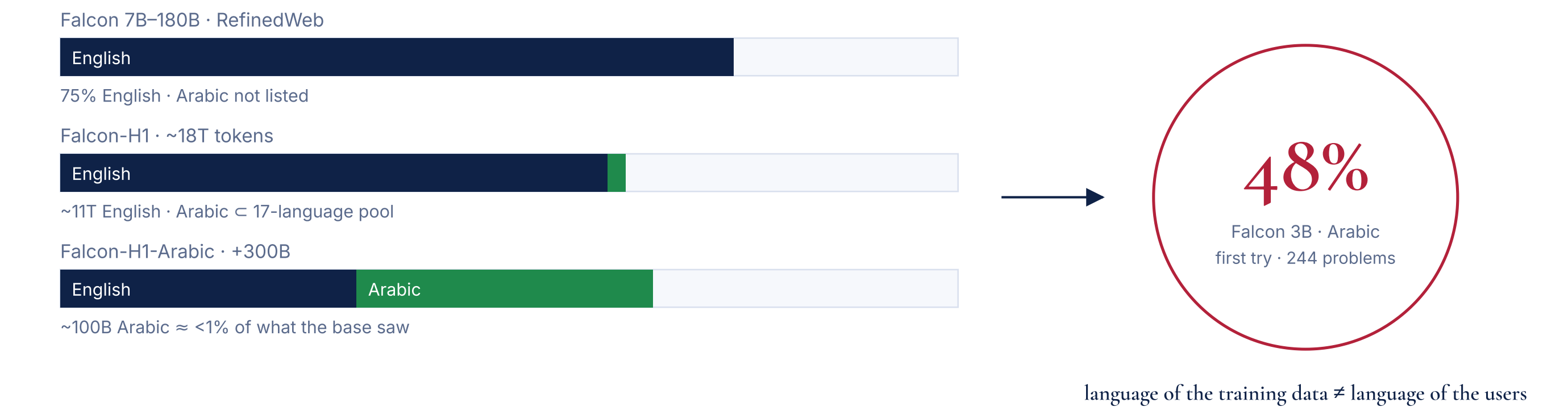


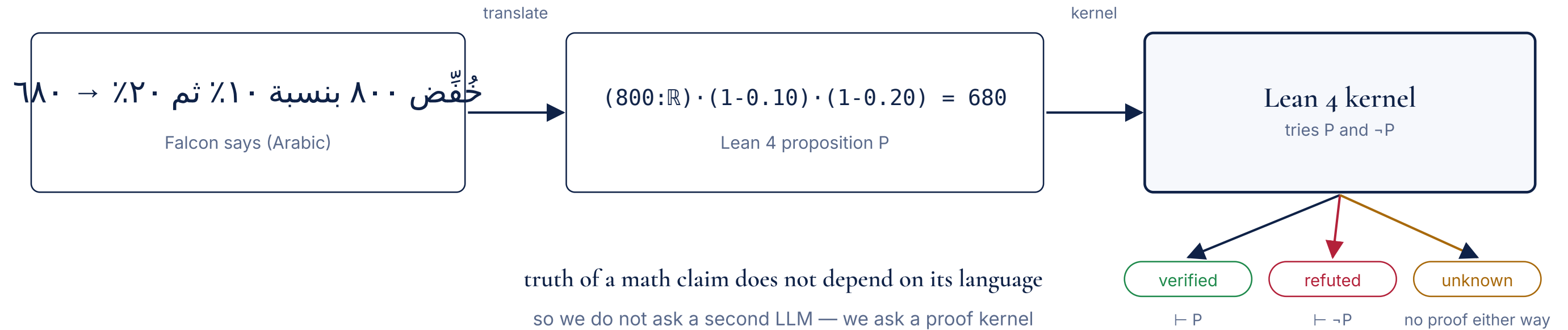
FalconVerifier

Falcon says it $\rightarrow$ Lean 4 proves it, or refutes it $\rightarrow$ Falcon learns. In Arabic.

Region's model · Arabic users · English data



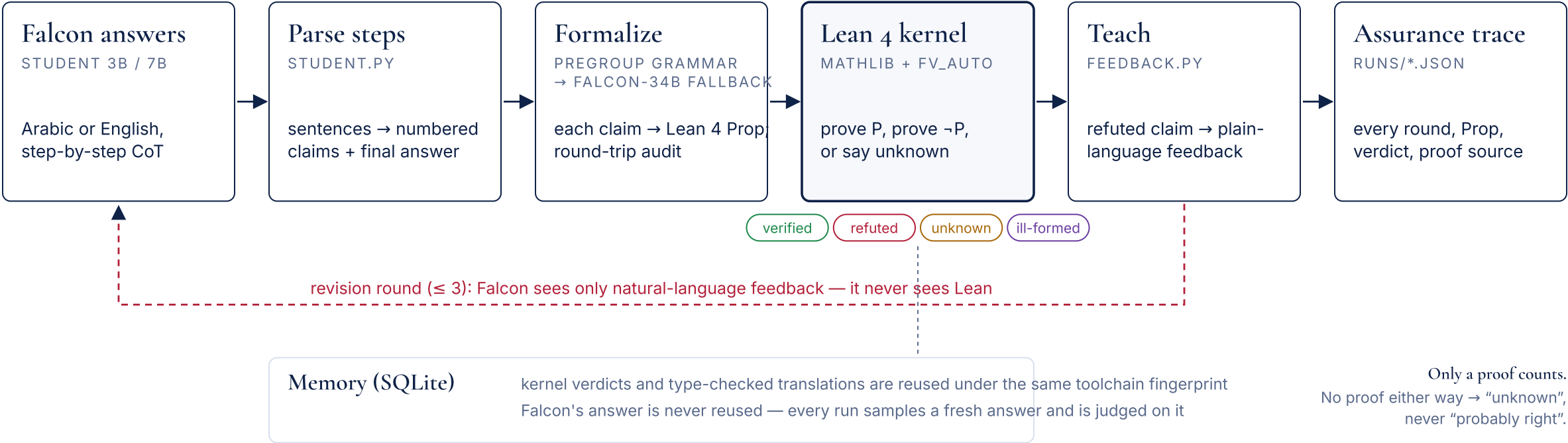
Don't ask another LLM. Ask a proof kernel.



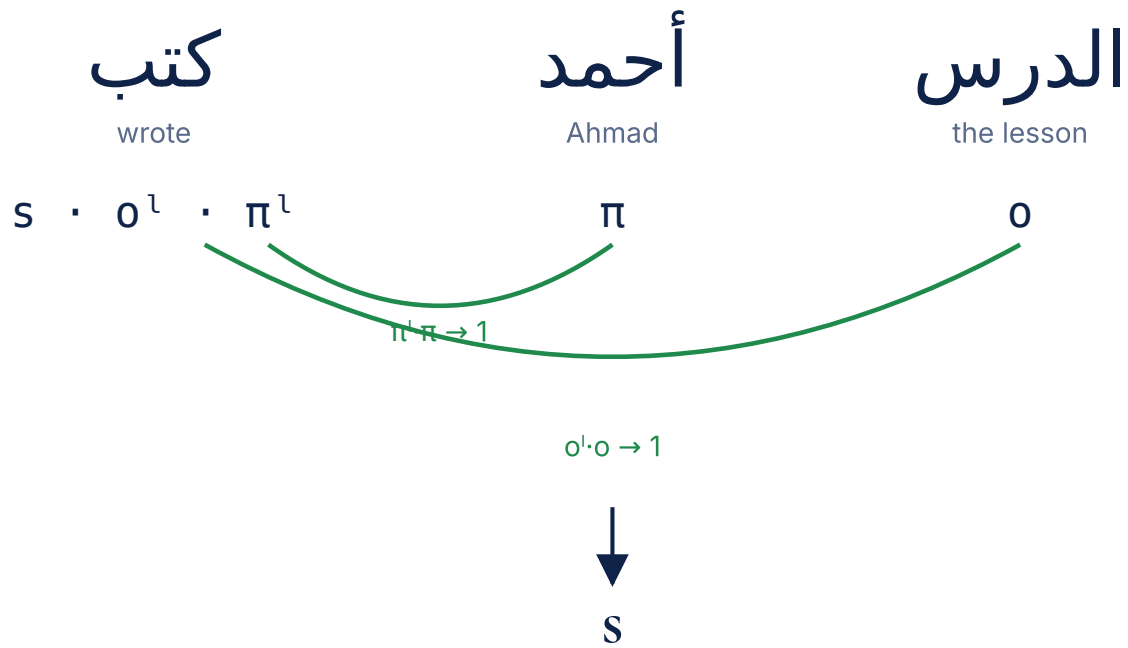
One loop, six stages

How a Falcon answer becomes a kernel-checked claim

FALCON SAYS IT · LEAN 4 DECIDES IT · FALCON LEARNS FROM IT



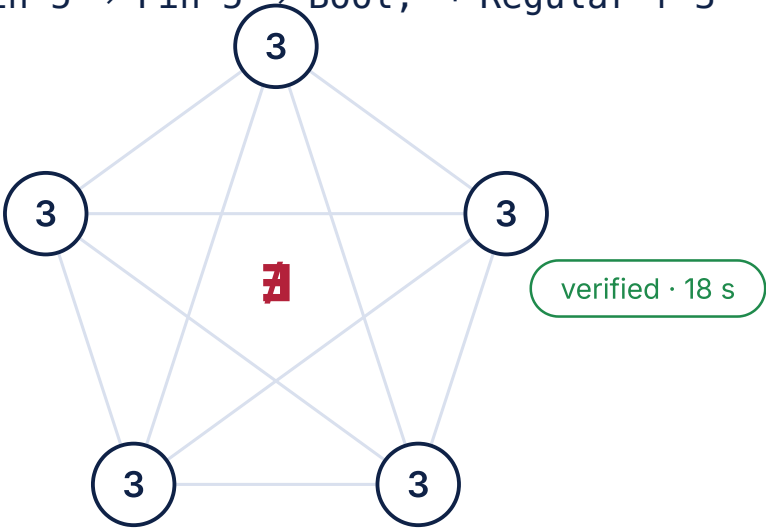
Pregroup grammar: types cancel $\Rightarrow$ translation is a proof



types cancel $\Rightarrow$ sentence $\Rightarrow$ the same derivation is the translation

أحمد كتب الدرس (SVO) — different word order, same meaning: proved in Lean (vso_svo_same_meaning)

$\forall f : \text{Fin } 5 \rightarrow \text{Fin } 5 \rightarrow \text{Bool}, \neg \text{Regular } f$



each claims 3 friends $\Rightarrow \sum \text{deg} = 5 \times 3 = 15$ (odd)

handshake lemma: $\sum \text{deg} = 2 \cdot |E|$ is even $\Rightarrow$ impossible $\Rightarrow$ someone lies ✓

Two examples, one screen

1 · catch → teach → correct

٢٨٠ %هاتف بـ 800 درهم، خُفِّض 10٪ ثم 20

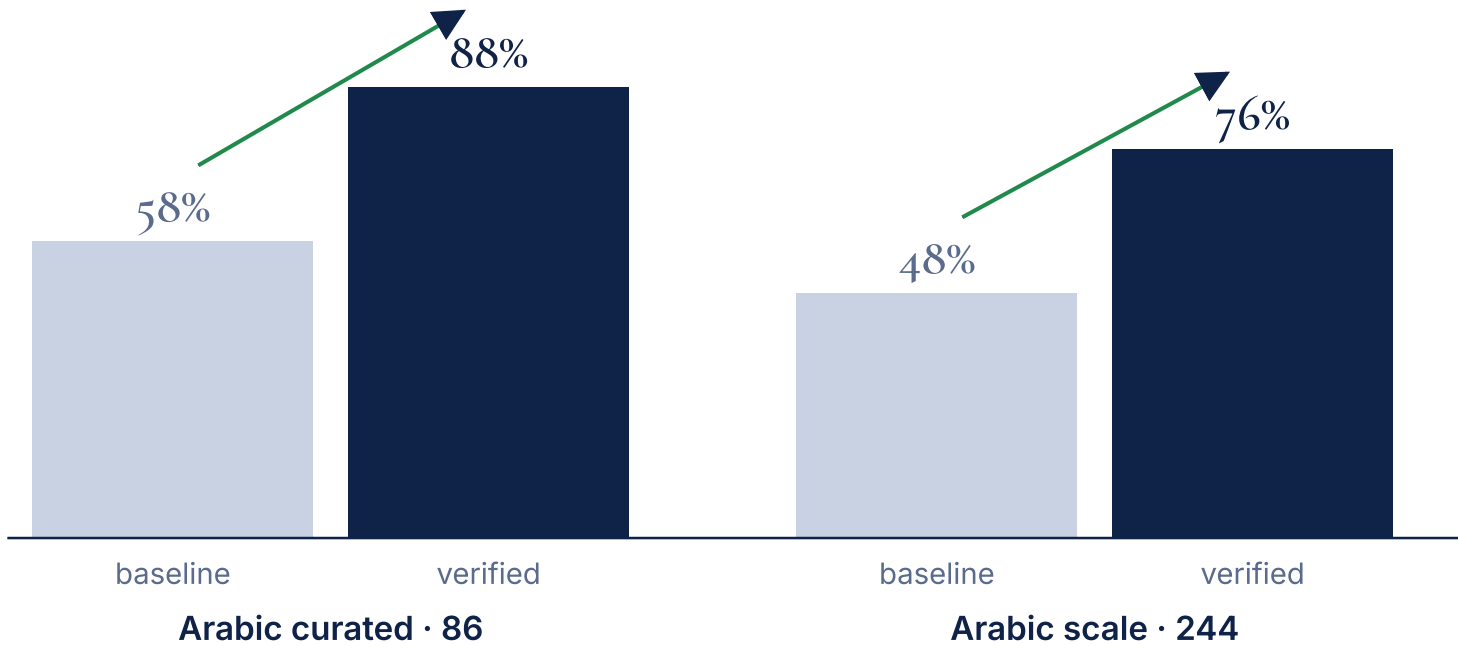


2 · prove without understanding

خمسة طلاب، كل واحد له ثلاثة أصدقاء — هل يكذب أحدهم؟



Falcon verifies Falcon — 0 regressions

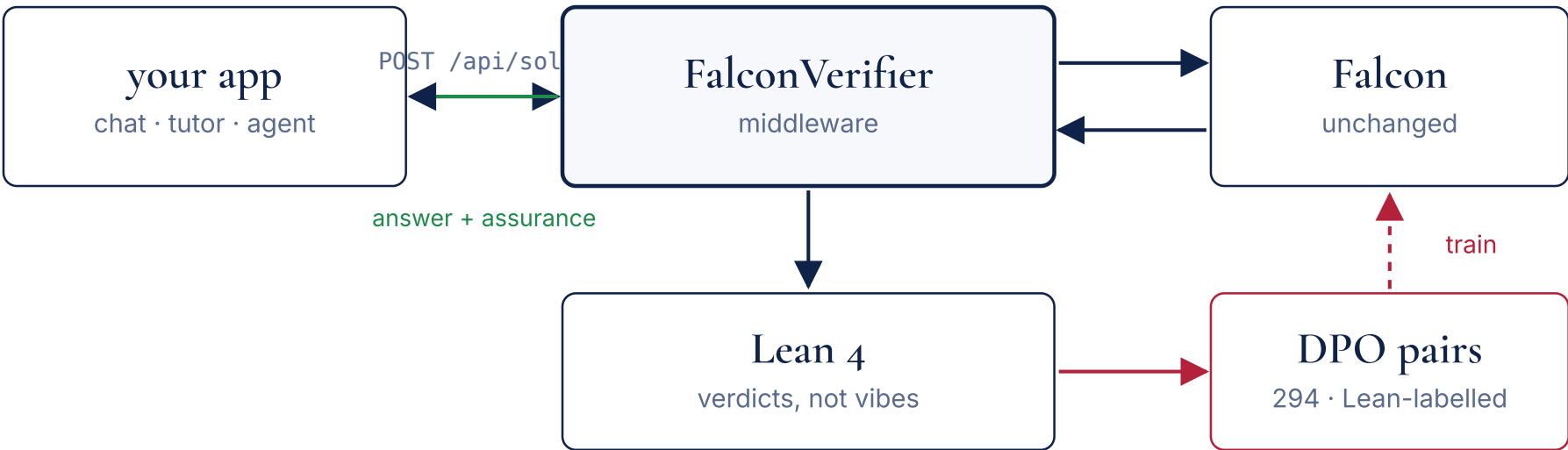


92%	wrong answers caught by Lean
69/127	fixed after one Arabic feedback
0	correct answers made wrong
1	false alarm in 244

3B student · 34B Arabic formalizer · Lean 4 + Mathlib

every bar links to its problems, traces and Lean sources on /benchmark

Middleware, not a model



- 📦 Docker · Lean + Mathlib inside
- 🔒 HTTPS · rate limits · access token
- 🔄 memory · repeated claims in 0 s
- 🔗 web UI · CLI · API · open source

limits, said out loud: Lean checks what was formalized · out-of-fragment steps depend on the 34B formalizer · proof-style problems → unknown

Trust in Arabic, proved in Lean.

Falcon $\rightarrow$ Lean 4 $\rightarrow$ Falcon

github.com/hw4375-collab/FalconVerifier · [live demo](#) $\rightarrow$